



DWF_dp_rnd functions

Arithmetic Rounding

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Description

Revision History

The DWF_dp_rnd functions truncate the lower bits of argument *a* below the bit position specified by argument *lsb* and return a rounded value according to the rounding mode specified by argument *mode*. Argument *a* and the return value are both either signed (two's complement) or unsigned.

Table 1-1 Function Names

Function Name	Description
DWF_dp_rnd	VHDL unsigned rounding
DWF_dp_rnd	VHDL signed (two's complement) rounding
DWF_dp_rnd_uns	Verilog unsigned rounding
DWF_dp_rnd_tc	Verilog signed (two's complement) rounding

Table 1-2 Argument Description

Argument Name	Type	Width / Values	Description
<i>a</i>	Vector	width	Input data
<i>lsb</i>	Integer	$\geq 1, \leq \text{width} - 1$	LSB index of return value (VHDL only, constant)
<i>mode</i>	Vector	4	Rounding mode (values in Table 1-4 on page 2)
DWF_dp_rnd	Vector	width - <i>lsb</i>	Return value

Table 1-3 Parameter Description (Verilog)

Parameter	Values	Description
<i>width</i>	≥ 1	Word length of input <i>a</i>
<i>lsb</i>	$\geq 1, \leq \text{width} - 1$	LSB index of return value

Verilog Include File: `DW_dp_rnd_function.inc`

Functional Description

Input $a[\text{width}-1:0]$ is divided into an integer part $a[\text{width}-1:\text{lsb}]$ and a fractional part $a[\text{lsb}-1:0]$ with the property

$$a[\text{width}-1:\text{lsb}] \leq a[\text{width}-1:\text{lsb}] . a[\text{lsb}-1:0] < a[\text{width}-1:\text{lsb}] + 1$$

Rounding either returns the integer part ($a[\text{width}-1:\text{lsb}]$) or its increment ($a[\text{width}-1:\text{lsb}] + 1$) depending on the fractional part ($a[\text{lsb}-1:0]$) and a threshold defined by the rounding mode.

$$z[\text{width}-\text{lsb}-1:0] = \text{DWF_dp_rnd}(a[\text{width}-1:0], \text{lsb}, \text{mode})$$

Effectively, the following occurs:

```
if a[lsb-1:0] > threshold(mode)
    z[width-lsb-1:0] = a[width-1:lsb]+1
else
    z[width-lsb-1:0] = a[width-1:lsb]
```

Rounding Modes

The DWF_dp_rnd functions provide a comprehensive set of rounding modes that are summarized in [Table 1-4](#). The rounding mode is selected by assigning argument `mode` either a predefined constant value or a 4-bit binary vector (dynamic).

The rounding modes differ with respect to rounding direction and tie resolution (a tie is the value $a[\text{width}+1:\text{lsb}].1$ that lies exactly in the middle between $a[\text{width}-1:\text{lsb}]$ and $a[\text{width}-1:\text{lsb}] + 1$). Rounding modes can be classified in terms of rounding symmetry, error bounds, bias (= accumulated error over many random samples) and hardware complexity, as shown in [Table 1-5](#) on page 3.

Table 1-4 Rounding Modes

Predefined Constant ^a	Value ^b	Rounding Direction	Tie Resolution	Class
DW_dp_rnd_neg_inf ^c	0000	Towards neg. infinity	None	A
DW_dp_rnd_pos_inf ^c	0001	Towards pos. infinity	None	B
DW_dp_rnd_zero ^c	0010	Towards zero	None	C (signed) A (unsigned)
DW_dp_rnd_away	0011	Away from zero	None	C (signed) B (unsigned)
DW_dp_rnd_even	0100	Towards even	None	C
DW_dp_rnd_odd	0101	Towards odd	None	C
DW_dp_rnd_near_neg_inf	1000	Towards nearest	Towards neg. infinity	D
DW_dp_rnd_near_pos_inf	1001	Towards nearest	Towards pos. infinity	E
DW_dp_rnd_near_zero	1010	Towards nearest	Towards zero	F (signed) D (unsigned)

Table 1-4 Rounding Modes (Continued)

Predefined Constant ^a	Value ^b	Rounding Direction	Tie Resolution	Class
DW_dp_rnd_near_away	1011	Towards nearest	Away from zero	F (signed) E (unsigned)
DW_dp_rnd_near_even ^c	1100	Towards nearest	Towards even	F
DW_dp_rnd_near_odd	1101	Towards nearest	Towards odd	F

a. Predefined constant value to be assigned to argument mode.

b. Binary vector to be assigned to argument mode. The individual bits can be interpreted as follows:

- mode[3] : nearest : 0 = not towards nearest, 1 = towards nearest
- mode[2:1] : symmetric : 00 = not symmetric, 01 = symmetric wrt zero, 10 = symmetric wrt even
- mode[0] : direction : 0 = towards negative/0/even, 1 = towards positive/infinity/odd

c. IEEE rounding modes (correspond to floor, ceil, trunc and round in the C programming language).

Table 1-5 Rounding Classes

Class	Symmetric	Error ^a	Bias	Complexity
A	no	$0 \leq \text{error} \leq 1 - 2^{-\text{lsb}}$	$(1 - 2^{-\text{lsb}})/2$	lowest
B	no	$-(1 - 2^{-\text{lsb}}) \leq \text{error} \leq 0$	$-(1 - 2^{-\text{lsb}})/2$	low
C	yes	$-(1 - 2^{-\text{lsb}}) \leq \text{error} \leq 1 - 2^{-\text{lsb}}$	0	highest
D	no	$-(1/2 - 2^{-\text{lsb}}) \leq \text{error} \leq 1/2$	$2^{-\text{lsb}}/2$	low
E	no	$-1/2 \leq \text{error} \leq 1/2 - 2^{-\text{lsb}}$	$-2^{-\text{lsb}}/2$	low
F	yes	$-1/2 \leq \text{error} \leq 1/2$	0	high

a. The error given is relative to the rounded output.

For more information about the DesignWare datapath functions, refer to the topic titled [Arithmetic – Datapath Functions Overview](#).

Related Topics

- [Arithmetic – Datapath Functions Overview](#)
- [DesignWare Building Block IP User Guide](#)

VHDL Example

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.numeric_std.all;
use DWARE.DW_dp_functions.all;
-- DWARE.DW_dp_functions_arith package if IEEE.std_logic_arith is used

entity DWF_dp_rnd_test is
  port (a, b, c : in  signed(7 downto 0);
        z       : out signed(7 downto 0));
end DWF_dp_rnd_test;

architecture rtl of DWF_dp_rnd_test is
begin
  z <= DWF_dp_rnd (a * b, 8, DW_dp_rnd_near_even) + c;
end rtl;
```

Verilog Example

```
module DWF_dp_rnd_test (a, b, c, z);

  input  signed [7:0] a, b, c;
  output signed [7:0] z;

  // Passes the parameters to the function
  parameter width = 16;
  parameter lsb   = 8;

  // add "$SYNOPSISYS/dw/sim_ver" to the search path for simulation
  `include "DW_dp_rnd_function.inc"

  assign z = DWF_dp_rnd_tc (a * b, DW_dp_rnd_near_even) + c;

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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